

Machine Learning – Question Bank

MODULE 1: Introduction & Understanding Data – 1

1. Explain the need for Machine Learning and compare it with traditional programming.
2. Describe the types of Machine Learning with suitable real-world examples.
3. Explain the Machine Learning process/lifecycle in detail.
4. Discuss the challenges of Machine Learning and possible mitigation strategies.
5. Explain univariate data analysis and descriptive statistics with examples and visualizations.

MODULE 2: Understanding Data – 2 & Basic Learning Theory

1. Differentiate between bivariate and multivariate data with examples.
2. Explain multivariate statistics and their role in Machine Learning.
3. Discuss the importance of feature engineering in building ML models.
4. Explain dimensionality reduction techniques and their significance.
5. Describe the design of a learning system and the concept of learning in Machine Learning.

MODULE 3: Similarity-Based Learning, Regression & Decision Trees

1. Explain the K-Nearest Neighbor (KNN) algorithm and its limitations.
2. Describe Weighted KNN and Locally Weighted Regression (LWR).
3. Explain linear and multiple linear regression, including assumptions.
4. Compare polynomial regression and logistic regression with use cases.
5. Explain the decision tree learning model and decision tree induction algorithms.

MODULE 4: Bayesian Learning & Artificial Neural Networks

1. Explain the fundamentals of Bayesian learning and Bayes' theorem.
2. Describe classification using the Naïve Bayes model.

3. Explain Bayesian learning for continuous attributes.
4. Describe the structure and working of an artificial neuron and perceptron learning algorithm.
5. Discuss types, applications, advantages, disadvantages, and challenges of Artificial Neural Networks.

MODULE 5: Clustering Algorithms & Reinforcement Learning

1. Explain different clustering approaches and proximity measures.
2. Describe hierarchical clustering and partition-based clustering algorithms.
3. Explain density-based and grid-based clustering approaches.
4. Discuss the components of Reinforcement Learning and its scope in Machine Learning.
5. Explain Markov Decision Process (MDP) and compare Q-learning and SARSA algorithms.